UET NOWSHERA | ELECTRICAL ENGINEERING

LAB REPORT 11

STRINGS IN C++

1. OBJECTIVE

- To understand the concept of strings as collections of characters.
- To differentiate between C-Strings (character arrays) and C++ string objects.
- To learn how to declare and initialize C-Strings.
- To utilize built-in functions for string manipulation (e.g., strcpy, strcat, strlen).

2. THEORETICAL BACKGROUND

C-String. In C programming, the collection of characters is stored in the form of arrays. This is also supported in C++ programming. That is why it is called C-strings. C-strings are arrays of type char terminated with a null character, that is, '\0' (ASCII value of null character is 0).

Declaring and Initializing. A C-String can be declared and initialized by providing a string literal, such as char myString[] = "Hello";. Although "Hello" has 5 characters, a null character is automatically added to the end. Unlike arrays, it is not necessary to use all the space allocated for the string.

String Object. In C++, you can also create a string object for holding strings. Unlike using char arrays, string objects have no fixed length, and can be extended as per your requirement. Using the standard C++ library string class provides a more robust and flexible way to handle text.

Reading Strings. When reading strings with spaces, the standard cin >> operator stops at the first space. To read an entire line of text including spaces, the cin.get() function is used for char arrays, and the getline() function is used for string objects.

3. SOFTWARE USED

- IDE: DEV-C++ IDE
- Compiler: C++ Compiler (bundled with DEV-C++)

4. EXPERIMENTAL TASKS

Task 1: String Length Calculation

A C++ program was created to declare and initialize a string variable with a specific phrase. The built-in length calculation function was then utilized to determine the number of characters in the string, which was subsequently displayed to the console.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    // Declare and initialize a string variable
    string message = "Welcome to the Electrical Engineering Department.";

    // Calculate and display the length of the string
    cout << "The message is: " << message << endl;
    cout << "The length of the string is: " << message.length() << "
characters." << endl;

    return 0;
}
```

Task 2: String Concatenation

Two separate string variables were declared and initialized representing a first name and a last name. These strings were then concatenated, with a space inserted between them, to form a complete full name. The final concatenated string was displayed.

```cpp
#include <iostream>
#include <string>
using namespace std;

int main() {
    // Declare two string variables
    string firstName = "Qazi";
    string lastName = "Ehsanullah";

    // Concatenate the strings to form a full name
    string fullName = firstName + " " + lastName;

    // Display the full name
    cout << "First Name: " << firstName << endl;
    cout << "Last Name: " << lastName << endl;
    cout << "Full Name: " << fullName << endl;

    return 0;
}
```

5. CONCLUSION

This lab focused on the implementation and manipulation of strings in C++. It highlighted the foundational differences between traditional C-style character arrays and modern C++ string objects. By practicing string initialization, length calculation, and concatenation, students learned to manage and process text-based data effectively. The use of string objects demonstrated a clear advantage in terms of flexibility and ease of use compared to fixed-length character arrays.